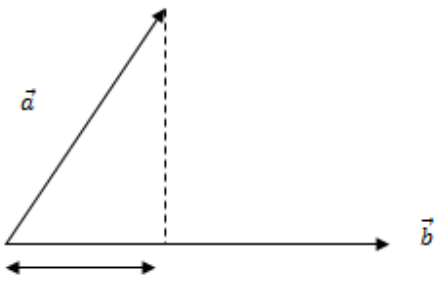




1. If $\vec{a} = 7\hat{i} + \hat{j} - 4\hat{k}$ and $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$ then find the projection of $\vec{a}$ on $\vec{b}$

Answer:

We know that projection of $\vec{a}$ on $\vec{b}$ means component of $\vec{a}$ along the direction of $\vec{b}$  $a \cos \theta = \text{Project of } \vec{a} \text{ on } \vec{b}$	We know $\vec{a} \cdot \vec{b} = ab \cos \theta$ Or $\frac{\vec{a} \cdot \vec{b}}{b} = a \cos \theta$ Or $\frac{(7\hat{i} + \hat{j} - 4\hat{k}) \cdot (2\hat{i} + 6\hat{j} + 3\hat{k})}{b} = a \cos \theta$ Or $\frac{14 + 6 - 12}{\sqrt{4 + 36 + 9}} = a \cos \theta$ Or $\frac{8}{7} = a \cos \theta$ Therefore Projection of $\vec{a}$ on $\vec{b} = \frac{8}{7}$
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2. Find λ , if the vectors $\vec{a} = \hat{i} + 3\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} - \hat{k}$ and $\vec{c} = \lambda\hat{j} + 3\hat{k}$ are coplanar

We know that if the vectors are coplanar then the volume of the parallelepiped formed by 3 vectors will be zero. Therefore for coplanar vector $\vec{a}, (\vec{b} \times \vec{c}) = 0$ $\begin{vmatrix} 1 & 3 & 1 \\ 2 & -1 & -1 \\ 0 & \lambda & 3 \end{vmatrix} = 0$	Or $1(-3 + \lambda) - 3(6 - 0) + 1(2\lambda - 0) = 0$ Or $-3 - 18 + 3\lambda = 0$ Or $\lambda = \frac{21}{3} = 7$ Answer $\lambda = 7$
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3. If the line makes angles 90° , 60° and θ with x,y and z-axis respectively, where θ is acute then find θ

We know that if a vector makes α, β and γ angle with X, Y and Z axis then

$$\cos \alpha = \frac{a_x}{\sqrt{a_x^2 + a_y^2 + a_z^2}}$$

$$\cos \beta = \frac{a_y}{\sqrt{a_x^2 + a_y^2 + a_z^2}}$$

$$\cos \gamma = \frac{a_z}{\sqrt{a_x^2 + a_y^2 + a_z^2}}$$

Therefore from above equation we have

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Given $\alpha = 90^\circ$, $\beta = 60^\circ$ and $\gamma = \theta (< 90^\circ)$

Therefore

$$\cos^2 90 + \cos^2 60 + \cos^2 \theta = 1$$

$$\text{Or } 0 + \frac{1}{4} + \cos^2 \theta = 1$$

$$\text{Or } \cos^2 \theta = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{Or } \cos \theta = \pm \frac{\sqrt{3}}{2} \text{ Since } \theta \text{ is acute}$$

$$\cos \theta = + \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ$$

Answer: $\theta = 30^\circ$

4. Write the element a_{23} of a 3×3 matrix $A = (a_{ij})$ whose elements a_{ij} are given by $a_{ij} = \frac{|i-j|}{2}$

Answer: Given $a_{ij} = \frac{|i-j|}{2}$

$$\text{Therefore } a_{23} = \frac{|2-3|}{2} = \frac{1}{2}$$



5. Find the differential equation representing the family of curves $V = \frac{A}{r} + B$, where A and B are arbitrary constants.

Answer: Given $V = \frac{A}{r} + B$

Differentiating with respect to r we get

$$\frac{dv}{dr} = -\frac{A}{r^2} + 0$$

$$\text{or } A = -r^2 \times \frac{dv}{dr}$$

Differentiating again with respect to r we get

$$0 = -\left(2r \frac{dv}{dr} + r^2 \frac{d^2v}{dr^2}\right)$$

$$\text{or } r^2 \frac{d^2v}{dr^2} + 2r \frac{dv}{dr} = 0$$

6. Find the integrating factor of the differential equation

$$\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right) \frac{dx}{dy} = 1$$

Answer:

We know for differential equation of the form $\frac{dy}{dx} + Py = Q \rightarrow (1)$ Integrating factor (IF) = $I = e^{\int P dx}$ Given equation $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right) \frac{dx}{dy} = 1$	or $\frac{dy}{dx} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}$ or $\frac{dy}{dx} + \frac{y}{\sqrt{x}} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$ Comparing with equation(1) : $P = \frac{1}{\sqrt{x}}$ Therefore Integrating factor (IF) = $I = e^{\int P dx}$ $I = e^{\int P dx} = e^{\int \frac{1}{\sqrt{x}} dx} = e^{2\sqrt{x}}$
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7. If $A = \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix}$ find $A^2 - 5A + 4I$ and hence find a matrix X such that $A^2 - 5A + 4I + X = 0$

Answer : $A^2 = \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix}$

$$= \begin{pmatrix} 2 \times 2 + 0 \times 2 + 1 \times 1 & 2 \times 0 + 0 \times 1 + 1 \times (-1) & 2 \times 1 + 0 \times 3 + 1 \times 0 \\ 2 \times 2 + 1 \times 2 + 3 \times 1 & 2 \times 0 + 1 \times 1 + 3 \times (-1) & 2 \times 1 + 1 \times 3 + 3 \times 0 \\ 1 \times 2 + (-1) \times 2 + 0 \times 1 & 1 \times 0 + (-1) \times 1 + 0 \times (-1) & 1 \times 1 + (-1) \times 3 + 0 \times 0 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{pmatrix}$$

$$A^2 - 5A + 4I = \begin{pmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{pmatrix} - 5 \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} + 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{pmatrix} - \begin{pmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{pmatrix} + \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{pmatrix} = -1 \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{pmatrix}$$

Therefore

$$A^2 - 5A + 4I = -1 \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{pmatrix}$$

Also we find $A^2 - 5A + 4I + \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{pmatrix} = 0$

Comparing with $A^2 - 5A + 4I + X = 0$

Answer

$$A^2 - 5A + 4I = -1 \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{pmatrix}$$



OR if $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix}$, find $(A')^{-1}$

Answer: $|A| = \begin{vmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{vmatrix} = 1(-1 - 8) - 2(0 + 8) + 3(0 - 2) = -9 + 16 - 6 = 1$

hence $|A| \neq 0$

We know that $A^{-1} = \frac{Adj.A}{|A|}$

To find Adj. A, let us find Cofactors of elements

$C_{11} = \begin{vmatrix} -1 & 4 \\ 2 & 1 \end{vmatrix} = -1 - 8 = -9$	$C_{12} = -\begin{vmatrix} 0 & 4 \\ -2 & 1 \end{vmatrix} = -8$	$C_{13} = \begin{vmatrix} 0 & -1 \\ -2 & 2 \end{vmatrix} = -2$
$C_{21} = -\begin{vmatrix} -2 & 3 \\ 2 & 1 \end{vmatrix} = 2 + 6 = 8$	$C_{22} = \begin{vmatrix} 1 & 3 \\ -2 & 1 \end{vmatrix} = 7$	$C_{23} = -\begin{vmatrix} 1 & -2 \\ -2 & 2 \end{vmatrix} = 2$
$C_{31} = \begin{vmatrix} -2 & 3 \\ -1 & 4 \end{vmatrix} = -8 + 3 = -5$	$C_{32} = -\begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} = -4$	$C_{33} = \begin{vmatrix} 1 & -2 \\ 0 & -1 \end{vmatrix} = -1$

We know that $A^{-1} = \frac{Adj.A}{|A|} = \frac{1}{1} \begin{bmatrix} -9 & 8 & -5 \\ -8 & 7 & -4 \\ -2 & 2 & -1 \end{bmatrix}$

Required $(A')^{-1} = (A^{-1})' = \begin{bmatrix} -9 & -8 & -2 \\ 8 & 7 & 2 \\ -5 & -4 & -1 \end{bmatrix}$



8. If $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, using properties of determinants find the value of $f(2x) - f(x)$

Answer: Multiplying R_2 by x

$$f(x) = \frac{1}{x} \begin{vmatrix} a & -1 & 0 \\ ax^2 & ax & -x \\ ax^2 & ax & a \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$

$$f(x) = \frac{1}{x} \begin{vmatrix} a & -1 & 0 \\ ax^2 & ax & -x \\ 0 & 0 & a+x \end{vmatrix}$$

$$\text{or } f(x) = \frac{1}{x} (a+x)(a^2x + ax^2)$$

$$\text{or } f(x) = a(a+x)(a+x)$$

$$\text{or } f(x) = a(a+x)^2 \rightarrow (1)$$

$$\text{Also } f(2x) = a(a+2x)^2 \rightarrow (2)$$

$$f(2x) - f(x) = a(a+2x)^2 - a(a+x)^2$$

$$= a(a^2 + 4x^2 + 4ax - a^2 - x^2 - 2ax)$$

$$= a(3x^2 + 2ax)$$



9. Find : $\int \frac{dx}{\sin x + \sin 2x}$

Answer: $I = \int \frac{dx}{\sin x + \sin 2x} = \int \frac{dx}{\sin x + 2\sin x \cos x} = \int \frac{dx}{\sin x(1 + 2\cos x)} = \int \frac{\sin x dx}{\sin^2 x(1 + 2\cos x)} = \int \frac{\sin x dx}{(1 - \cos^2 x)(1 + 2\cos x)} =$

or $I = \int \frac{\sin x dx}{(1 - \cos x)(1 + \cos x)(1 + 2\cos x)}$

Putting $\cos x = u$, $-\sin x dx = du$

$I = \int \frac{-du}{(1 - u)(1 + u)(1 + 2u)}$

Let $\frac{1}{(1 - u)(1 + u)(1 + 2u)} = \frac{A}{1 - u} + \frac{B}{1 + u} + \frac{C}{1 + 2u}$

or $1 = A(1 + u)(1 + 2u) + B(1 - u)(1 + 2u) + C(1 - u)(1 + u)$

Putting $u = 1$, $A = \frac{1}{6}$

Putting $u = -1$, $B = -\frac{1}{2}$

Putting $u = -\frac{1}{2}$, $C = \frac{4}{3}$

$\frac{1}{(1 - u)(1 + u)(1 + 2u)} = \frac{1}{6(1 - u)} - \frac{1}{2(1 + u)} + \frac{4}{3(1 + 2u)}$

$I = \int -\left[\frac{1}{6(1 - u)} - \frac{1}{2(1 + u)} + \frac{4}{3(1 + 2u)}\right] du$

or $I = \frac{1}{6} \log|1 - u| + \frac{1}{2} \log|1 + u| - \frac{2}{3} \times \log|1 + 2u| + C$

or $I = \frac{1}{6} \log|1 - \cos x| + \frac{1}{2} \log|1 + \cos x| - \frac{2}{3} \times \log|1 + 2\cos x| + C$



OR Integrate the following with respect to $x \frac{x^2-3x+1}{\sqrt{1-x^2}}$

Answer: $I = \int \frac{x^2-3x+1}{\sqrt{1-x^2}} dx$

$$\frac{x^2 - 3x + 1}{\sqrt{1 - x^2}} = -\frac{-x^2 + 3x - 1}{\sqrt{1 - x^2}} = -\frac{1 - x^2 + 3x - 2}{\sqrt{1 - x^2}} = -\left(\frac{1 - x^2}{\sqrt{1 - x^2}} + \frac{3x}{\sqrt{1 - x^2}} - \frac{2}{\sqrt{1 - x^2}}\right)$$

$$I = -\left(\int \sqrt{1 - x^2} dx - \frac{3}{2} \int \frac{-2x}{\sqrt{1 - x^2}} dx - 2 \int \frac{1}{\sqrt{1 - x^2}} dx\right)$$

$$I = -\left(I_1 - \frac{3\sqrt{1 - x^2}}{2 \times \frac{1}{2}} - 2\sin^{-1}x\right) + C_1 \rightarrow (1)$$

where $I_1 = \int \sqrt{1 - x^2} dx$

Putting $x = \sin\theta, dx = \cos\theta d\theta$

or $I_1 = \int \cos\theta \times \cos\theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta = \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta = \frac{1}{2}\sin^{-1}x + \frac{1}{4} \times 2x\sqrt{1 - x^2} + C_2 \rightarrow (2)$

Putting (2) in (1) we get

$$I = -\left(\frac{1}{2}\sin^{-1}x + \frac{1}{4} \times 2x\sqrt{1 - x^2} + C_2 - \frac{3\sqrt{1 - x^2}}{2 \times \frac{1}{2}} - 2\sin^{-1}x\right) + C_1$$

$$I = -\frac{1}{2}\sin^{-1}x - \frac{1}{4} \times 2x\sqrt{1 - x^2} + \frac{3\sqrt{1 - x^2}}{2 \times \frac{1}{2}} + 2\sin^{-1}x + C$$

$$I = \frac{3}{2}\sin^{-1}x + 3\sqrt{1 - x^2} - \frac{x}{2}\sqrt{1 - x^2} + C$$



10. Evaluate: $\int_{-\pi}^{\pi} (\cos ax - \sin bx)^2 dx$

Answer: $I = \int_{-\pi}^{\pi} (\cos ax - \sin bx)^2 dx$

$$\text{or } I = \int_{-\pi}^{\pi} (\cos^2 ax + \sin^2 bx + 2\cos ax \sin bx) dx$$

$$\text{or } \int_{-\pi}^{\pi} \left(\frac{1 + \cos 2ax}{2} + \frac{1 - \cos 2bx}{2} + 2\cos ax \sin bx \right) dx$$

$$\text{or } I = \int_{-\pi}^{\pi} dx + \frac{1}{2} \int_{-\pi}^{\pi} \cos 2ax dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos 2bx dx + 2 \int_{-\pi}^{\pi} \cos ax \sin bx dx$$

$$\text{or } I = [x]_{-\pi}^{\pi} + \frac{1}{2} \left[\frac{\sin 2ax}{2a} \right]_{-\pi}^{\pi} - \frac{1}{2} \left[\frac{\sin 2bx}{2b} \right]_{-\pi}^{\pi} + 2I_1$$

$$I = \{\pi - (-\pi)\} + \frac{1}{2 \times 2a} (0 - 0) - \frac{1}{2 \times 2b} (0 - 0) + 2I_1$$

$$\text{or } I = 2\pi + 2I_1 \rightarrow (1)$$

$$\text{Where } I_1 = \int_{-\pi}^{\pi} \cos ax \sin bx dx$$

Let $g(x) = \cos ax \sin bx$ then

$$g(-x) = \cos a(-x) \sin b(-x) = -\cos ax \sin bx = -g(x)$$

Thus $g(x)$ is odd function.

$$\text{Therefore } I_1 = \int_{-\pi}^{\pi} \cos ax \sin bx dx = 0$$

Putting in equation (1) we get

$$I = 2\pi$$